Lesson Planning

Complete Lesson Plan Book

Class: Class 12

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Matrices: Addition, Subtraction and Scalar Multiplication

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Learning Objectives

• Students will state the order of a matrix and identify its elements.

• Students will add and subtract matrices of the same order.

• Students will multiply a matrix by a scalar and simplify the result.

Key Concepts

• A matrix is a rectangular arrangement of numbers in rows and columns; a 2 by 3 matrix has 2 rows and 3 columns.

• Two matrices can be added or subtracted only when they have the same order, and the operation is done element by element.

• Scalar multiplication means multiplying every element of a matrix by the same number.

• Adding matrices follows the same rules as adding numbers: it is commutative and associative when orders match.

Teaching Methodology (Step by Step)

1. Write two 2 by 2 matrices on the board and ask students what the numbers in the first row and second column mean.

2. Add the two matrices element by element, circling matching positions.

3. Show why a 2 by 2 matrix and a 2 by 3 matrix cannot be added.

4. Model scalar multiplication with a negative scalar, stressing the sign of every element.

5. Students solve five matrix operations in pairs and check each other's orders first.

Classroom Activities

• Matching orders: students sort matrix pairs into addable and non-addable groups.

• Sign check: students find the sign error in a worked scalar multiplication.

• Matrix relay: teams pass a matrix problem along, each student doing one step.

Assessment Questions

1. State the order of a matrix with 3 rows and 2 columns.

2. Add the matrices [[2, 5], [1, 4]] and [[3, 0], [2, 1]].

3. Multiply [[4, -2], [0, 3]] by -3.

Homework

• Find A + B and A - B for A = [[1, 7], [5, 2]] and B = [[4, 3], [0, 6]].

• Explain in three lines why matrices of different orders cannot be added.

Time Allocation

40 minutes